

Empirical Evaluation of Out-Of-Distribution Performance of Tabular Foundation Models

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Abstract. Tabular Foundation Models (TFMs) have emerged as novel approaches for tabular predictive tasks, demonstrating competitive predictive performance to ensemble tree-based models. Most TFMs are trained and evaluated on independent and identically distributed data, but this assumption changes in real-world scenarios due to distribution shifts, which compromise the robustness of models. Limited research has been conducted of TFMs under distribution shifts. We present an empirical evaluation of Out-Of-Distribution (OOD) performance of nine TFMs, spanning diverse pre-training strategies and architectures: TabPFNV2, TabPFNV2.5, TabPFNV2.6, TabPFNV3, TabICL, TabICLv2, Mitra, LimiX and TabFM. Four real-world datasets from the TableShift study were considered (HELOC, Voting, Childhood Lead, Diabetes Readmission), covering label, socioeconomic, and geographic shift types. Our results show that most evaluated TFMs degrade under distribution shift regardless of pre-training strategy, with shift gaps ranging from 0.009 to 0.045 depending on shift type and model. A small subset of synthetic-prior models (TabFM, TabICL, TabICL v2) showed marginally higher OOD than ID performance on HELOC. We also identified a scalability gap, as high-performing models demand significant memory and computational resources beyond what standard deployment infrastructure can support. This study extends existing benchmarks for OOD in tabular data, providing evidence to support their adoption in high-stakes domains characterized by structural distribution shifts.

Keywords: Tabular foundation models · distribution shifts · out-of-distribution · TabPFN · TabICL · Mitra · TabFM and LimiX

1 Introduction

Tabular data are among most extended data formats for representing structured information, which is characterized by a *table-like format*, where rows and

columns correspond to samples and features, respectively [7]. For years, tree-based ensemble models, including XGBoost, LightGBM, and CatBoost, have been the state-of-the-art in predictive tabular tasks due to their high performance and interpretability [15]. Although several attempts of using deep learning models for tabular data have been proposed (*e.g.*, TabNet [1], SAINT [14]), they have shown lower performance compared to tree-based models.

Tabular Foundation Models (TFMs) have emerged as novel approaches, showing remarkable generalization to downstream tasks without fine-tuning [7]. A remarkable type of TFMs, often referred to as Prior-Data Fitted Networks (PFNs), exploits in-context learning (ICL) at inference time, passing the labeled training subset as context and predicting test samples without gradient updates. These TFMs have been trained using diverse collections of tabular data, including synthetic and real-world datasets, and or mixtures of both [15]. In particular, TabPFN marked a milestone for tabular classification, outperforming tree-based ensemble models on small data [6]. Recent TFMs such as TabDPT [10], Mitra [18], TabICL [11], and Limix [17], have also outperformed tree-based models across different benchmarks. TFMs have been used in a range of applications such as disease classification [3, 8], intrusion detection [12, 13] among others.

Despite the success of TFMs, existing models are developed under the assumption that all samples of training and test subsets follow the same In-Distribution (ID). However, this assumption often breaks down in real-world applications, where deployed models are often exposed to unseen samples drawn from distributions that differ from the training data, known as Out-Of-Distribution (OOD). Such distributional shifts can significantly degrade model performance, resulting in inaccurate predictions of underrepresented domains [16]. While existing benchmarks (such as TableShift [4]) have evaluated the OOD effect on traditional machine learning models, the robustness of TFMs to distribution shifts remains underexplored.

In this paper, we conduct an empirical evaluation of the performance of TFMs under OOD scenarios. We evaluate nine TFMs with different pre-training strategies, including: *(i)* real-world data pre-training (TabPFN v2, TabPFN v2.5, TabPFN v2.6, TabPFN v3); *(ii)* mixed prior structural causal models with tree ensemble priors (Mitra); and *(iii)* synthetic priors (TabICL, TabICL v2, LimiX, TabFM). We evaluate on four TableShift datasets, covering label shift (HELOC), socioeconomic shift (Childhood Lead), geographic shift (Voting) and admission-source shift (Diabetes Readmission). These were selected to cover three application domains (healthcare, finance and public health). To our knowledge, this is the first work to investigate the impact of OOD and distribution shifts on the predictive performance of TFMs. The main contributions of this study are:

- We provide the first empirical evaluation of TFMs under distribution shifts, covering four real-world datasets, and four shift types.
- We evaluate the impact of pre-training approach (with synthetic, real-world, or mixed data) of TFMs on OOD robustness.
- We extend the OOD analysis of tabular predictive models into TFMs, assessing the relationship between ID and OOD performance.

- We provide evidence of the computational constraints of TFMs under dataset sizes, finding a practical implication for model deployment.
- We provide empirical evidence of TFMs under distribution shifts across healthcare and financial domains.

2 Materials and methods

2.1 Datasets

We evaluate four TFMs on datasets from TableShift [4], selected to cover a range of domains and distribution shift types. Table 1 summarizes the datasets considered. The remaining TableShift datasets were not included in this study for two main reasons. First, some of them (ICU Length of Stay and ICU Mortality) requires credentialed access to the MIMIC-III database via PhysioNet, involving human-subjects research training and a formal data use agreement. Second, most of datasets range from approximately 125,000 to nearly 6 million samples, substantially exceeding the in-context window and inference-time memory budget of the TFMs evaluated.

Table 1: A summary of datasets used in this study.

Dataset	Shift Variable	Features	Train	Validation	ID test	OOD test
Diabetes readmission	Admission source	183	34288	4286	4287	50968
Childhood Lead	Poverty level	17	11807	1476	1476	11466
HELOC	Label distribution	38	2220	278	278	6914
Voting	Geographic region	380	37548	4693	4694	23103

Diabetes Readmission comprises hospital records of patients with diabetes, with the prediction target indicating whether a patient was readmitted within 30 days of discharge. Distributional shift is defined according to the admission source: referral admissions constitute the ID subset, whereas emergency admissions constitute the OOD subset. Childhood Lead is derived from the National Health and Nutrition Examination Survey. The task is to predict whether a child has elevated blood lead levels. The domain split is defined by household poverty level, introducing a socioeconomic subpopulation shift between ID and OOD. HELOC is a financial risk dataset, where the task is to predict whether a consumer will repay a home equity line of credit. The distribution shift in this dataset is characterized by label shift, *e.g.*, the marginal distribution of the target variable differs substantially between the ID and OOD subsets. Voting is constructed from the American National Election Studies Time Series Cumulative Data File, covering presidential elections from 1948 to 2020. The task is to predict voter turnout. The domain split is defined by U.S. Census region, with the Southern states constituting the OOD subset and other regions forming the training distribution. This geographic shift reflects a real-world scenario in which

a model trained on one regional population is applied to another with different political and demographic characteristics.

2.2 Tabular foundation models

TFMs comprise a broad group of approaches for transferring knowledge across tabular prediction tasks. A prominent line within this group is formed by PFNs and related ICL-based TFMs, which perform inference without the need for parameter updates and provide competitive predictive performance on small-scale datasets. TFMs present differences in pre-training approaches. Some models are pre-trained on collections of real-world tabular datasets, aiming to learn statistical patterns that generalize across different domains. In contrast, other models are pre-trained exclusively on synthetic datasets generated from Structural Causal Models (SCMs) (or probabilistic generative processes), with no exposure to real-world tabular data.

TabPFN v1 uses a transformer architecture to perform ICL on embeddings of rows. It was pre-trained on millions of datasets generated from SCMs to approximate Bayesian inference over tabular classifiers [6]. This enables the model to learn transferable causal inductive biases that generalize to unseen real-world data. The original TabPFN was designed for datasets with a maximum context window of 1,000 training rows and 100 numerical features, exhibiting limited performance on categorical data [2]. To address scalability, several variants have been proposed. TabPFN v2 includes dense token subsampling and improved task conditioning, supporting up to 10,000 samples [9]. TabPFN v2, v2.5, and v2.6 uses a transformer architecture that alternates row-wise and feature-wise attention, enhancing predictive performance but introducing computational complexity that scales poorly with dataset size [5].

TabPFN v3 introduces several changes in the architecture to improve scalability. For classification, it employs an attention-based retrieval decoder instead of a fixed-width multilayer perceptron output head, allowing it to handle an arbitrary number of classes via soft nearest-neighbor retrieval [5]. It introduces row chunking, enabling fixed-size row processing that decouples GPU memory usage from dataset size while preserving the predictions of full inference. Also, it leverages multi-query attention to minimize key-value cache memory consumption during inference [5].

TabFM is a zero-shot ICL-based TFM developed by Google Research that presents the strengths of both TabPFN and TabICL using a novel architecture built with three mechanisms: *(i)* alternating row-and-column attention by capturing complex feature interactions without manual feature engineering; *(ii)* row compression that condenses each row’s cross-attended representation into a single dense vector; and *(iii)* a dedicated ICL transformer that operates over compressed row embeddings. In contrast to the TabPFN variants, TabFM is trained on hundreds of millions of synthetic datasets generated using SCMs.

TabICL addresses the scalability bottleneck of TFMs by replacing the joint attention with a two-stage architecture, reducing complexity and scaling to large number of samples [11]. Pre-trained on synthetic data generated from SCMs,

TabICL achieves competitive performance while requiring less memory and lower computational resources than TabPFN variants. TabICLv2 introduces three key improvements over TabICL: *(i)* a novel synthetic data generation engine for high pretraining diversity; *(ii)* architectural enhancements (including a scalable softmax); and *(iii)* a new pre-training approach by replacing AdamW with the Muon optimizer [11]. Mitra demonstrates that mixed priors combining SCMs and tree ensembles effectively model decision boundary geometry [18]. In contrast to TabPFNV2 and TabICLv2, which merge each feature and its label into a single token, Mitra represents the label using a separate token for each row. LimiX represents tabular data as a set-valued sample–feature entity rather than collapsing each row into a fixed embedding, enabling it to model dependencies across both samples and features [17].

3 Results

3.1 Experimental setup

TFMs are used in their default inference configurations without task-specific fine-tuning, and employing the default setup in the original papers. For each dataset we use the three splits defined by TableShift: a training split used as the in-context examples for TFM inference, an ID test split drawn from the same distribution as training, and an OOD test split drawn from the held-out domain. No validation data is used for hyperparameter tuning, as all TFMs are evaluated in their default configurations.

We report the Area Under the Receiver Operating Characteristic (ROC-AUC) values for each model-dataset combination. Due to the class imbalance present in several datasets, ROC-AUC is considered as the evaluation metric. The shift gap (Δ ROC-AUC) is defined for each metric as the difference between ID and OOD performance, where larger positive values indicate greater degradation under shift. All experiments are repeated across five random seeds. We report mean and standard deviation across seeds. Due to the high inference computational cost of TabFM and Limix for the Voting and Diabetes Readmission datasets, results are not reported.

3.2 Classification results

Figure 1 and Table 2 present the ID and OOD ROC-AUC for all TFMs and the datasets considered. OOD performance was lower than ID performance for the majority of model-dataset combinations, confirming that distribution shift degrades predictive results for most TFMs. Three synthetic-prior models (TabFM, TabICL, TabICL v2) showed a small negative shift gap on HELOC, achieving higher OOD than ID ROC-AUC. Models based on real-world pre-training (TabPFN variants) consistently achieved the highest performance both ID and OOD across all four datasets. The shift gap varies substantially by dataset and shift type: mean shift gap was largest for Childhood Lead (socioeconomic shift,

mean $\Delta \approx 0.036$), followed by Diabetes Readmission (admission-source shift, mean $\Delta \approx 0.020$), then Voting (geographic shift, mean $\Delta \approx 0.004$), with HELOC (label shift, mean $\Delta \approx 0.001$) showing the smallest average gap. Mitra performed competitively on Voting dataset, but showed the largest degradation on HELOC and Childhood Lead, being less robust to distribution shifts. The gap between ID and OOD persists across all pre-training strategies and all shift types evaluated.

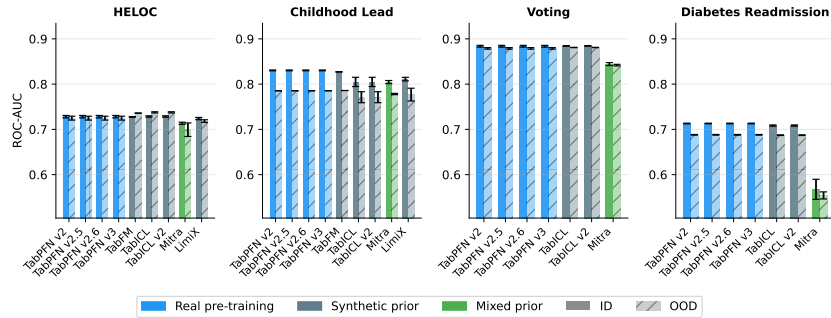


Fig. 1: In-distribution and out-of-distribution results for all models.

Figure 2 shows the ID versus OOD ROC-AUC values for each model across the datasets, with the dashed diagonal representing the no-degradation reference. Most points lie below the diagonal, confirming that the majority of models suffer performance degradation under distribution shift. However, on HELOC, TabFM, TabICL, and TabICL v2 fall slightly above the diagonal, indicating marginally better OOD than ID performance for these synthetic-prior models on this dataset. On HELOC (label shift), models place in a narrow ID range (0.70–0.74) but diverge more on OOD performance: TabICL and TabICL v2 achieved the highest OOD scores despite moderate ID performance, while Mitra shows the largest drop, indicating high sensitivity to label shift. LimiX also falls below other models in OOD performance relative to its ID ROC-AUC value.

On Childhood Lead (socioeconomic shift), the TabPFN variants (v2, v3) occupies the top-right region (highest ID and OOD), while synthetic-prior models (TabICL, TabICL v2) place at lower ID performance and suffer a larger gap from the diagonal. Mitra and LimiX again fall below most models in OOD performance. On Voting (geographic shift), all models except Mitra achieved high ID performance (0.88–0.90) and the shift gap is visually small, with most points lying close to the diagonal. TabICL and TabICL v2 are competitive with the TabPFN variants. Across all panels, the distance below the diagonal is largest for Childhood Lead, followed by Diabetes Readmission, while Voting and HELOC show the smallest degradation. This contrasts with the original TableShift study, which identified label shift as one of the main drivers of OOD degradation

Table 2: In-distribution and out-of-distribution ROC-AUC and shift gap Δ (mean $\pm$ std over 5 seeds) for all models and datasets. ‘-’ indicates the model-dataset combination was not evaluated due to high inference computational cost.

Model	HELOC			Childhood Lead		
	ID	OOD	Δ	ID	OOD	Δ
TabPFN v2	0.728 $\pm$ 0.003	0.725 $\pm$ 0.004	0.003 $\pm$ 0.004	0.830 $\pm$ 0.001	0.785 $\pm$ 0.000	0.045 $\pm$ 0.001
TabPFN v2.5	0.728 $\pm$ 0.003	0.725 $\pm$ 0.004	0.003 $\pm$ 0.004	0.830 $\pm$ 0.001	0.785 $\pm$ 0.000	0.045 $\pm$ 0.001
TabPFN v2.6	0.728 $\pm$ 0.003	0.725 $\pm$ 0.004	0.003 $\pm$ 0.004	0.830 $\pm$ 0.001	0.785 $\pm$ 0.000	0.045 $\pm$ 0.001
TabPFN v3	0.728 $\pm$ 0.003	0.725 $\pm$ 0.004	0.003 $\pm$ 0.004	0.830 $\pm$ 0.001	0.785 $\pm$ 0.000	0.045 $\pm$ 0.001
TabFM	0.728 $\pm$ 0.001	0.736 $\pm$ 0.000	-0.008 $\pm$ 0.001	0.827 $\pm$ 0.000	0.786 $\pm$ 0.000	0.041 $\pm$ 0.000
TabICL	0.729 $\pm$ 0.002	0.738 $\pm$ 0.001	-0.009 $\pm$ 0.002	0.805 $\pm$ 0.010	0.771 $\pm$ 0.012	0.034 $\pm$ 0.017
TabICL v2	0.729 $\pm$ 0.002	0.738 $\pm$ 0.001	-0.009 $\pm$ 0.002	0.805 $\pm$ 0.010	0.771 $\pm$ 0.012	0.034 $\pm$ 0.017
Mitra	0.713 $\pm$ 0.002	0.699 $\pm$ 0.015	0.014 $\pm$ 0.015	0.804 $\pm$ 0.004	0.778 $\pm$ 0.001	0.026 $\pm$ 0.002
LimiX	0.724 $\pm$ 0.003	0.718 $\pm$ 0.003	0.006 $\pm$ 0.004	0.811 $\pm$ 0.004	0.777 $\pm$ 0.014	0.034 $\pm$ 0.015

Model	Voting			Diabetes Readmission		
	ID	OOD	Δ	ID	OOD	Δ
TabPFN v2	0.884 $\pm$ 0.002	0.879 $\pm$ 0.002	0.005 $\pm$ 0.000	0.713 $\pm$ 0.001	0.688 $\pm$ 0.001	0.025 $\pm$ 0.001
TabPFN v2.5	0.884 $\pm$ 0.002	0.879 $\pm$ 0.002	0.005 $\pm$ 0.000	0.713 $\pm$ 0.001	0.688 $\pm$ 0.001	0.025 $\pm$ 0.001
TabPFN v2.6	0.884 $\pm$ 0.002	0.879 $\pm$ 0.002	0.005 $\pm$ 0.000	0.713 $\pm$ 0.001	0.688 $\pm$ 0.001	0.025 $\pm$ 0.001
TabPFN v3	0.884 $\pm$ 0.002	0.879 $\pm$ 0.002	0.005 $\pm$ 0.000	0.713 $\pm$ 0.001	0.688 $\pm$ 0.001	0.025 $\pm$ 0.001
TabFM	-	-	-	-	-	-
TabICL	0.884 $\pm$ 0.001	0.881 $\pm$ 0.000	0.004 $\pm$ 0.001	0.708 $\pm$ 0.001	0.687 $\pm$ 0.001	0.021 $\pm$ 0.001
TabICL v2	0.884 $\pm$ 0.001	0.881 $\pm$ 0.000	0.004 $\pm$ 0.001	0.708 $\pm$ 0.001	0.687 $\pm$ 0.001	0.021 $\pm$ 0.001
Mitra	0.844 $\pm$ 0.003	0.842 $\pm$ 0.002	0.002 $\pm$ 0.003	0.568 $\pm$ 0.022	0.554 $\pm$ 0.008	0.013 $\pm$ 0.017
LimiX	-	-	-	-	-	-

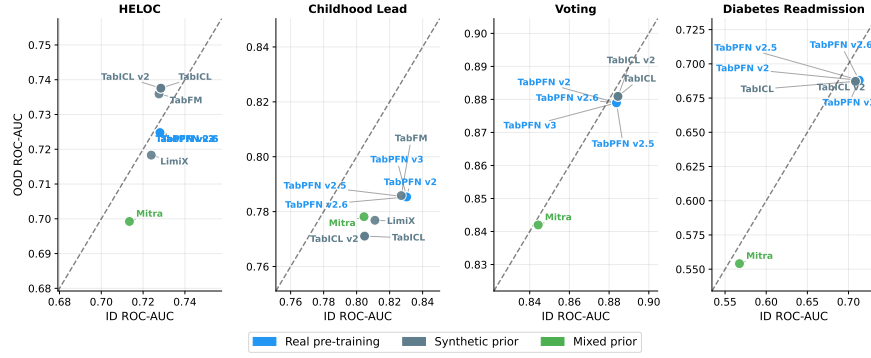


Fig. 2: In-distribution versus out-of-distribution ROC-AUC scatter plots per dataset. Each point represents one model, coloured by pre-training strategy (blue: real-world, slate: synthetic, green: mixed). The dashed diagonal line represents no degradation under shift, points below the diagonal indicate OOD performance worse than ID.

for traditional models. In our TFM evaluation, label shift (HELOC) shows the least degradation on average.

Figure 3 compares OOD ROC-AUC and Shift Gap Δ across the three pre-training corpus strategies: Real, Mixed, and Synthetic. The left panel revealed that real-data pre-training achieved the highest median and mean OOD performance (median ≈ 0.78 , mean ≈ 0.79), followed by Mixed (median ≈ 0.775) and Synthetic (median ≈ 0.77), though the differences in central tendency are modest. Notably, the Real group exhibits the widest interquartile range, reflecting the diversity of TabPFN variants (v2 through v3), which span a broader performance spectrum across datasets.

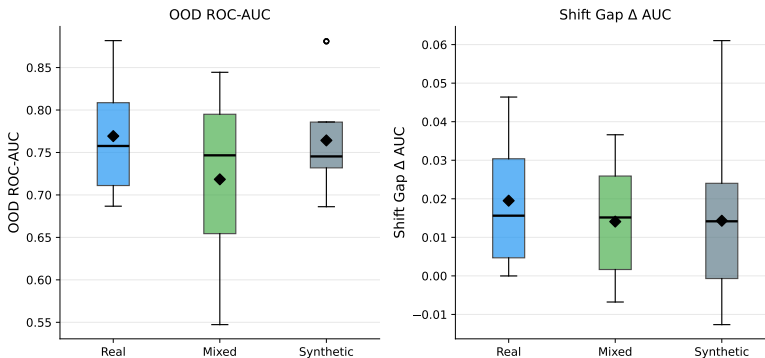


Fig. 3: Pre-training corpus effect on out-of-distribution robustness. Left panel: distribution of out-of-distribution ROC-AUC by pre-training strategy. Right panel: distribution of shift gap ROC-AUC.

In the right panel, the Real group shows the largest mean shift gap (mean ≈ 0.019), indicating that real pre-training models can suffer substantial degradation under certain shift types. The Mixed and Synthetic groups both exhibit lower mean shift gaps (mean ≈ 0.015 and ≈ 0.012 , respectively), with Synthetic showing the lowest median gap of the three groups. Both Mixed and Synthetic include negative gap values, meaning some models in these groups marginally improve under OOD conditions on certain dataset. These findings suggest that pre-training corpus type is not the primary factor of OOD robustness. While real pre-training models tend to achieve higher absolute OOD performance, they do not exhibit smaller shift gaps, indicating that the performance advantage observed in Figure 1 is driven by stronger ID performance rather than by superior distributional robustness.

4 Discussion

This study presented a comprehensive evaluation of TFMs under distribution shifts, covering different TFMs across real-world TableShift datasets spanning

label shift (HELOC), socioeconomic shift (Childhood Lead), geographic shift (Voting), and admission-source shift (Diabetes Readmission). Our results yield five main findings.

First, pre-training with real-world datasets does not provide OOD robustness. While models pre-trained on real-world data (TabPFN variants) achieved higher absolute OOD performance, their shift gaps are comparable to synthetic-prior models (TabICL, LimiX, TabFM). These results suggest that real-data pre-training does not improve robustness to distribution shifts. Second, it was revealed that relationship between ID and OOD performance documented by Gardner et al. [4] for traditional tabular models extends to TFMs. In all datasets, models with higher ID performance tend to achieve higher OOD performance, with no model consistently breaking above or below the proportional degradation line. The results indicate that improvements in ID performance, achieved through improved pre-training, larger context windows, or architectural refinements, are reflected proportionally under distribution shift. However, higher ID performance does not necessarily result in better OOD generalization, indicating that TFMs lack an inherent robustness advantage.

Third, the type of distribution shift emerged as a key determinant of OOD degradation. Across all models, Childhood Lead, characterized by a socioeconomic subpopulation shift, exhibited the largest performance losses, followed by Diabetes Readmission, which involves an admission-source shift. In contrast, HELOC, characterized by label shift, and Voting, characterized by geographic shift, showed modest degradation. HELOC exhibited the smallest average ID–OOD performance gap among the four datasets, with several synthetic-prior models even achieving marginal improvements in OOD performance. This ordering only partially agrees with findings reported in TableShift, where label shift is identified as an important factor associated with OOD degradation in traditional tabular models. Our results therefore suggest that the relationship between shift type and OOD robustness may differ for TFMs.

Fourth, Mitra demonstrates greater sensitivity to label shift than the other evaluated models. Although its performance remains competitive under socioeconomic and geographic shifts, it exhibits the largest OOD degradation on HELOC. This pattern suggests that its pre-training prior, which combines SCMs with tree ensembles, is well suited to covariate and subpopulation shifts but less robust to changes in the label distribution. These findings suggest that Mitra should be used with caution in environments where label prevalence may differ between training and deployment, such as clinical risk prediction across healthcare systems with different patient populations.

Fifth, scalability remains a major practical limitation of TFMs. A recurring challenge was the inability of several TFMs to process large datasets without significant computational constraints. In particular, TabFM and LimiX exhibited high memory growth with respect to the number of training samples and features, making them impractical on standard hardware for datasets with medium size. LimiX’s cross-attention mechanism scales as $\mathcal{O}(n_{\text{train}} \times n_{\text{test}} \times d_{\text{model}})$, requiring chunked inference and training context subsampling even on a 47GB

GPU (NVIDIA RTX 6000 Ada) for the Childhood Lead dataset (11,807 training rows, 17 features) and making experiments with Voting dataset (37,548 rows, 380 features) infeasible without subsampling. Similarly, both LimiX and TabFM were computationally infeasible on the Diabetes Readmission dataset (34,288 training rows, 183 features) without substantial subsampling. TabFM additionally required chunking on OOD test sets to avoid out-of-memory errors, despite being explicitly designed for scalability. Mitra imposed a hard limit of 10,000 training rows due to quadratic memory growth, a constraint that affected three of the four datasets considered in this study. These results reveal a fundamental trade-off: the highest-performing TFMs are often the least scalable, limiting their deployment to environments with high-end GPU resources and specialized memory-management techniques. Existing benchmarks obscure this limitation because they evaluate models on relatively small datasets that remain well within the memory capacity of all methods.

5 Conclusions

This paper presented the first OOD evaluation of modern TFMs, evaluating nine TFMs spanning synthetic, mixed, and real-world pre-training strategies. We evaluated on four TableShift datasets covering label, socioeconomic, and geographic shifts. Our results showed that TFMs do not exhibit inherent robustness to distribution shift. The performance degrades systematically across all models, shift types, and pre-training strategies, and the relationship between ID and OOD performance documented for traditional models extends to TFMs. Pre-training corpus in TFMs does not provide OOD robustness. Although models pre-trained on real-world data achieve superior OOD performance, they show shift gaps similar to those of synthetic-prior models, suggesting that their advantage is driven by stronger ID fitting rather than an inherent capacity for OOD generalization. This work extends OOD benchmarks for tabular data and provides evidence supporting their application in high-stakes scenarios involving structural distribution shifts.

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References

1. Arik, S.Ö., Pfister, T.: Tabnet: Attentive interpretable tabular learning. In: Proceedings of the AAAI conference on artificial intelligence. vol. 35, pp. 6679–6687 (2021)
2. Behre, O.: In-context forecasting in supply chains: Evaluating the promise and limits of tabular foundation models (2026)

3. Chushig-Muzo, D., Calero-Díaz, H., Fabelo, H., Årsand, E., van Dijk, P.R., Soguero-Ruiz, C.: Characterizing the impact of physical activity on patients with type 1 diabetes using statistical and machine learning models. *Applied Sciences* **14**(21), 9870 (2024)
4. Gardner, J., Popovic, Z., Schmidt, L.: Benchmarking distribution shift in tabular data with tableshift. *Advances in Neural Information Processing Systems* **36**, 53385–53432 (2023)
5. Grinsztajn, L., Flöge, K., Key, O., Birkel, F., Jund, P., Roof, B., Manium, M., Hoo, S.B., Bühler, M., Garg, A., et al.: Tabpfn-3: Technical report. arXiv preprint arXiv:2605.13986 (2026)
6. Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S.B., Schirmeister, R.T., Hutter, F.: Accurate predictions on small data with a tabular foundation model. *Nature* **637**(8045), 319–326 (2025)
7. Jiang, J.P., Liu, S.Y., Cai, H.R., Zhou, Q.L., Ye, H.J.: Representation learning for tabular data: A comprehensive survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026)
8. Lara-Abelenda, F.J., Chushig-Muzo, D., Wäagner, A.M., Tayefi, M., Soguero-Ruiz, C.: Interpretable and multimodal fusion methodology to predict severe hypoglycemia in adults with type 1 diabetes. *Engineering Applications of Artificial Intelligence* **144**, 110142 (2025)
9. Lee, K.: Multitabpfn: Codebook-based extensions of tabpfn for high-class-count tabular classification. *Neural Networks* p. 108932 (2026)
10. Ma, J., Thomas, V., Hosseinzadeh, R., Labach, A., Kamkari, H., Cresswell, J.C., Golestan, K., Yu, G., Caterini, A.L., Volkovs, M.: Tabdpt: Scaling tabular foundation models on real data. arXiv preprint arXiv:2410.18164 (2024)
11. Qu, J., Holzmüller, D., Varoquaux, G., Morvan, M.L.: Tabiclv2: A better, faster, scalable, and open tabular foundation model. arXiv preprint arXiv:2602.11139 (2026)
12. Ruiz-Villafranca, S., Roldán-Gómez, J., Carrillo-Mondejar, J., Martinez, J.L., Gañán, C.H.: Wfe-tab: Overcoming limitations of tabpfn in iiot-mec environments with a weighted fusion ensemble-tabpfn model for improved ids performance. *Future Generation Computer Systems* **166**, 107707 (2025)
13. Ruiz-Villafranca, S., Roldán-Gómez, J., Gómez, J.M.C., Carrillo-Mondéjar, J., Martinez, J.L.: A tabpfn-based intrusion detection system for the industrial internet of things: S. ruiz-villafranca et al. *The Journal of Supercomputing* **80**(14), 20080–20117 (2024)
14. Somepalli, G., Goldblum, M., Schwarzschild, A., Bruss, C.B., Goldstein, T.: Saint: Improved neural networks for tabular data via row attention and contrastive pre-training. arXiv preprint arXiv:2106.01342 (2021)
15. Somvanshi, S., Das, S., Javed, S., Antariksa, G., Hossain, A.: A survey on tabular data: from tree-based methods to tabular deep learning. *ACM Computing Surveys* (2026)
16. Yang, J., Zhou, K., Li, Y., Liu, Z.: Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision* **132**(12), 5635–5662 (2024)
17. Zhang, X., Ren, G., Yu, H., Yuan, H., Wang, H., Li, J., Wu, J., Mo, L., Mao, L., Hao, M., et al.: Limix: Unleashing structured-data modeling capability for generalist intelligence. arXiv preprint arXiv:2509.03505 (2025)
18. Zhang, X., Maddix Robinson, D., Yin, J., Erickson, N., Ansari, A.F., Han, B., Zhang, S., Akoglu, L., Faloutsos, C., Mahoney, M., et al.: Mitra: Mixed synthetic priors for enhancing tabular foundation models. *Advances in neural information processing systems* **38**, 15795–15840 (2026)